07 · Did the regional campaign work? — synthetic control (Anchor B)

July 12, 2026

The business decision. We ran a TV campaign in one metro area — a **DMA** (*Designated Market Area*, the standard US media-buying region) — and left the rest of the country dark. Sales in that metro are up since launch. Before we spend to roll the campaign out nationally: **was the lift caused by the campaign, or would sales have risen there anyway?**

0.0.1 Why this is hard: there was no control group

Unlike the uplift notebook (01), here there is **no randomized hold-out** — we didn't flip a coin to decide which metro got the campaign, we just picked one. So we can't simply compare it to "the others." And we can't just compare the treated metro's sales *after* vs *before*, because sales everywhere are pushed around by a shared **trend** (secular growth), **seasonality**, and a common **macro wave** (the economy, category demand). A naive before/after would credit the campaign with an uplift the market would have delivered on its own. We need the one thing we can never observe: **what sales in that metro *would have been* if we had not run the campaign** — its *counterfactual*.

0.0.2 The idea: build the counterfactual from the other markets

Synthetic control (Abadie & Gardeazabal, 2003) constructs that missing counterfactual as a **weighted blend of the untreated markets** — the "**donor pool**." It searches for a set of weights so that the blend closely tracks the treated market's sales *in the pre-launch period*. If a weighted combination of, say, 60% Denver + 25% Phoenix + 15% others matched our metro week-for-week *before* the campaign, then — assuming those markets keep moving together — that same blend is our best guess of what our metro *would* have done *after*, absent the campaign. The **gap** between the metro's actual sales and this "synthetic" version, after launch, is the estimated lift.

The reason it can work at all: all markets share the same underlying **factors** (trend, season, macro). A blend of donors that matches the treated unit's pre-period is, in effect, matching it on those shared factors — so the blend and the treated unit stay locked together *unless something (the campaign) breaks them apart.*

We do this the Bayesian way, so instead of one number we get a **posterior** — a full distribution of plausible lifts — which gives us honest uncertainty, a real significance test (a **placebo permutation**, defined in Depth B), and robustness checks a VP of Growth can interrogate before committing national budget.

0.0.3 How this notebook is organised

The **7-step contract** (question · simulate a known truth · identify · estimate · validate · decide in € · caveats) plus the full **three depths**:

- **Depth A** — why the tempting naive estimators fail (a bake-off)
- **Depth B** — placebo inference in *space* and *time*, with the scale-free Abadie RMSE-ratio test
- **Depth C** — the euro rollout decision and its sensitivity to assumptions

```
FAST=True  SC sampling={'draws': 600, 'tune': 600, 'chains': 2}
```

0.1 2 · Simulate a ground truth (why fake data first)

As in every notebook here, we **cannot grade a causal method on real data** — the counterfactual ("what our metro would have done without the campaign") is never observed. So we build a simulator where we *plant* the true lift, confirm the method recovers it, and only then trust it.

The simulator produces a weekly **sales panel**: a table with one row per market per week, for 30 DMAs over 60 weeks. Each market's sales are built from **shared latent factors** — a rising trend, a seasonal cycle, and a random macro wave common to all markets — combined with that market's own baseline level, its own sensitivity to each factor ("factor loadings"), and idiosyncratic noise. The campaign launches in **week 40** in market 0 only, adding a **true +12% lift** from then on.

The shared macro wave is the villain: it moves *every* market together, so it's exactly what a naive before/after would mistake for campaign effect. But it's also the hero of the method — because the factors are *shared*, a weighted blend of the *other* markets can reconstruct market 0's untreated path. That is precisely what makes the effect *identifiable* here (recoverable in principle), and the plot below shows the setup: one dark line (treated) inside a bundle of grey donors, all riding the same wave.

The data-generating model — exactly what `dgp.geo_panel` implements (defaults & seed in `src/cmp/dgp.py`). Markets $i = 0, \dots, 29$, weeks $t = 0, \dots, 59$. Three **shared latent factors**:

$$\begin{aligned} g_t &= 0.4\,t && \text{(trend)},\\ s_t &= 8\sin\!\left(\tfrac{2\pi t}{26}\right) + 4\sin\!\left(\tfrac{2\pi t}{13}\right) && \text{(season)},\\ m_t &= \sum_{u\le t} \eta_u,\quad \eta_u \sim \mathcal{N}(0,\,1.2^2) && \text{(macro random walk)}. \end{aligned}$$

Each market has its own baseline $c_i \sim U(80, 140)$ and factor loadings $\lambda_i = (\lambda_{i1}, \lambda_{i2}, \lambda_{i3})$, each $\sim U(0.6, 1.4)$:

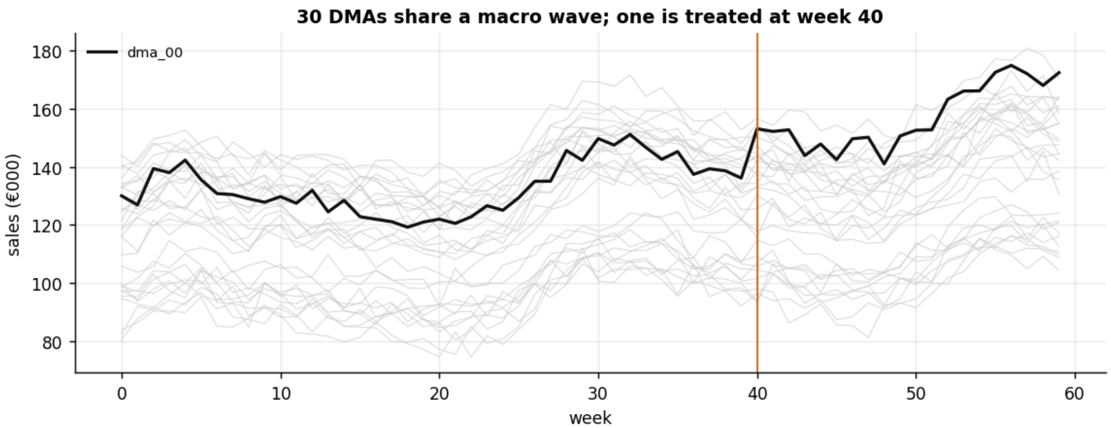
$$Y_{it} = c_i + \lambda_{i1}\,g_t + \lambda_{i2}\,s_t + \lambda_{i3}\,m_t + \varepsilon_{it}, \qquad \varepsilon_{it} \sim \mathcal{N}(0,\,3^2),$$

and the campaign adds a **proportional lift to market 0 only**, from week 40:

$$Y_{0t} = c_0 + \lambda_{01}\,g_t + \lambda_{02}\,s_t + \lambda_{03}\,m_t + \varepsilon_{0t} + \Delta_t, \qquad \Delta_t = 0.12\,(c_0 + \lambda_{01}\,g_t + \lambda_{02}\,s_t + \lambda_{03}\,m_t)\,\mathbf{1}[t \ge 40].$$

This factor structure is precisely why synthetic control works here: every market rides the *same* (g_t, s_t, m_t), only with different loadings, so a convex combination of donors whose blended loadings match $(\lambda_{01}, \lambda_{02}, \lambda_{03})$ reproduces market 0's untreated path. It's also why the naive estimators fail — the random-walk m_t moves all markets together, and a before/after comparison books that shared drift as "lift."

```
Treated: dma_00, launch week 40.  TRUE lift €17.0k/week, total €339k.
```



0.2 3 · Identify — the estimand and the assumptions it rests on

The estimand (the precise thing we *want*). For each post-launch week t, the causal effect on the treated market is the potential-outcome contrast

$$\tau_t \;=\; Y_{1,t}(1) \;-\; Y_{1,t}(0), \qquad t > T_0,$$

where $Y_{1,t}(1)$ is **observed** (the campaign ran) but $Y_{1,t}(0)$ — what week-t sales *would have been* with no campaign — is the **missing counterfactual**.

The estimator. Synthetic control fills in that counterfactual with a weighted average of donor markets, so the effect we actually compute is

$$\hat{\tau}_t \;=\; Y_{1,t} \;-\; \underbrace{\textstyle\sum_j w_j\, Y_{j,t}}_{\widehat{Y_{1,t}(0)} \;=\; \text{synthetic control}}, \qquad t > T_0,$$

where $Y_{1,t}$ is the **treated** market's sales in week t, $Y_{j,t}$ is **donor** market j's sales, T_0 is the launch week, and w_j are the **donor weights**. The weights are chosen to match the treated market's *pre-launch* path and are constrained to the **simplex**: each $w_j \geq 0$ and they sum to 1 ($\sum_j w_j = 1$). Think of the simplex constraint as "the synthetic market must be a genuine *weighted average* of real markets" — no negative weights, no scaling beyond the observed range.

Why the simplex matters (the core of Abadie's method). If instead we let weights be any numbers (ordinary regression), the "synthetic" market could sit *outside* the range of any real market — **extrapolation** dressed up as a match, which fits the pre-period beautifully and then diverges wildly after. Forcing weights onto the simplex keeps the synthetic inside the **convex hull**

of the donors (loosely, “inside the cloud of real markets”), which is what makes the post-launch projection credible.

The assumptions — each gets an explicit check later:

assumption	plain meaning	checked in
convex hull / no extrapolation	the treated market lies inside the donor cloud; simplex weights enforce it	weights (Step 4)
good pre-fit	the synthetic actually tracks the treated market *before* launch	pre-RMSE gate (Step 5)
no anticipation	nothing shifts sales *before* launch (no leaked campaign, no pre-buying)	placebo-in-time (Depth B)
stable factor structure	donors and treated keep co-moving *after* launch (the loadings matched pre-launch still hold)	long pre-period + placebo-in-time
no spillover / interference	the campaign in market 0 doesn’t change donor markets (SUTVA for geos)	stress-tested by simulation (Depth B): planted spillover *attenuates* the estimate, and excluding the contaminated donors restores it — placebo-in-space *cannot* detect spillover (it contaminates every placebo too), so on real data exclude adjacent / media-overlapping donors up front
no concurrent treated shock	nothing else hit market 0 exactly at launch	untestable — institutional knowledge only

0.3 4 · Estimate — a Bayesian synthetic control

We put a **Dirichlet prior** on the donor weights and fit them to the pre-period. (The *Dirichlet* is the natural probability distribution over the simplex — draws from it are automatically non-negative and sum to 1 — so it’s the Bayesian way to say “the weights live on the simplex.”) Fitting returns a **posterior** over the weights, and therefore a posterior over the whole counterfactual path and the lift — Abadie’s constrained least squares, but with uncertainty for free.

The fitted model, in symbols. With Y_{0t} the treated market and Y_{jt} the $J = 29$ donors, fit **on the pre-period only** ($t < 40$):

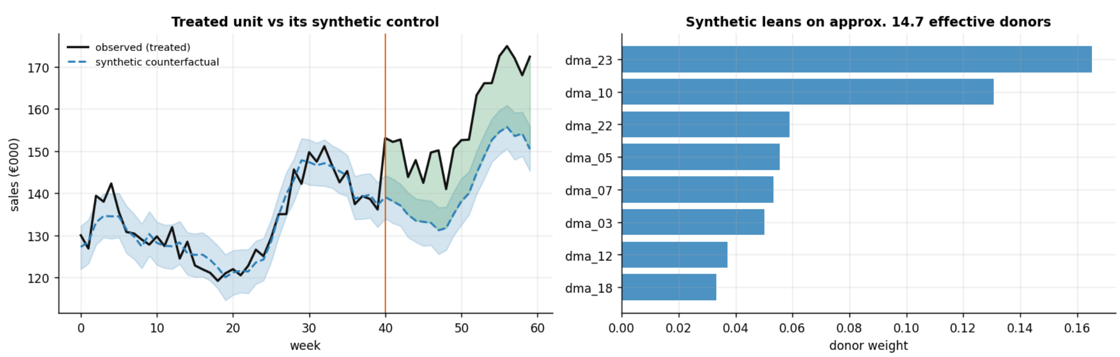
$$Y_{0t} \sim \mathcal{N}\Big(\textstyle\sum_{j=1}^{J} w_j\, Y_{jt},\ \ \sigma^2\Big), \qquad w \sim \mathrm{Dirichlet}(\mathbf{1}_J), \qquad \sigma \sim \mathrm{HalfNormal}(5).$$

$\mathrm{Dirichlet}(\mathbf{1}_J)$ is flat over the simplex — no donor favoured a priori, weights non-negative and summing to 1 by construction. The $\sigma \sim \mathrm{HalfNormal}(5)$ scale is likewise tied to the units of the data: sales are in €000s, weekly levels sit around €80–170k, and the DGP's idiosyncratic noise has sd €3k — so a prior sd scale of €5k is weakly informative on the residual: wide enough to cover any plausible blend error, tight enough to rule out absurd ones (a €50k weekly residual would mean the "synthetic" tracks nothing). The counterfactual for *every* week is then the posterior-predictive blend $\widehat{Y}^{(0)}_{0t} = \sum_j w_j Y_{jt} + \varepsilon_t$ with $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ — the ε_t matters: dropping it (using the mean blend alone) understates the uncertainty band, which is exactly the coverage lesson step 5 quantifies — and the weekly lift is $Y_{0t} - \widehat{Y}^{(0)}_{0t}$.

Two things to read off the fit: - **Which donors** the synthetic leans on (the weight bars). One caveat to state plainly: the Dirichlet prior produces a *regularized blend* spread over many donors, rather than classical synthetic control's sparse handful. That's **more stable** out-of-sample but **less individually interpretable**. We summarise the spread as an **"effective number of donors"** $= 1/\sum_j w_j^2$ (the inverse Herfindahl index — 1 if all weight is on one donor, J if spread evenly over J donors). - **Pre-fit RMSE** — the root-mean-square gap between the treated market and its synthetic *before* launch, in €000. Small means the synthetic is a good stand-in; this is the precondition for believing the post-launch gap — Step 5 turns this into a concrete PASS/FAIL gate (pre-RMSE below ~a third of the weekly lift it must detect).

> **On real data.** You don't need our simulator — this method runs on *your own* weekly (or daily) sales-by-market panel, which every company already has: one treated market, the rest as donors. The canonical *public* example is Abadie's **California Proposition 99** tobacco study (California vs 38 other US states); it has the identical shape (one treated unit, many donors, a known intervention date) and is the standard dataset to reproduce this on.

```
SC sampling convergence: max r-hat 1.010 - min ESS 836 - divergences 0
pre-fit RMSE €2.86k  ·  estimated total lift €305k (true €339k)  ·  90% CI
[€272k, €342k] - at full precision the truth sits inside this band (§5's multi-
seed check explains why the total-lift band runs optimistic)
```



0.4 5 · Validate — recovery, calibration, the pre-fit gate, and robustness

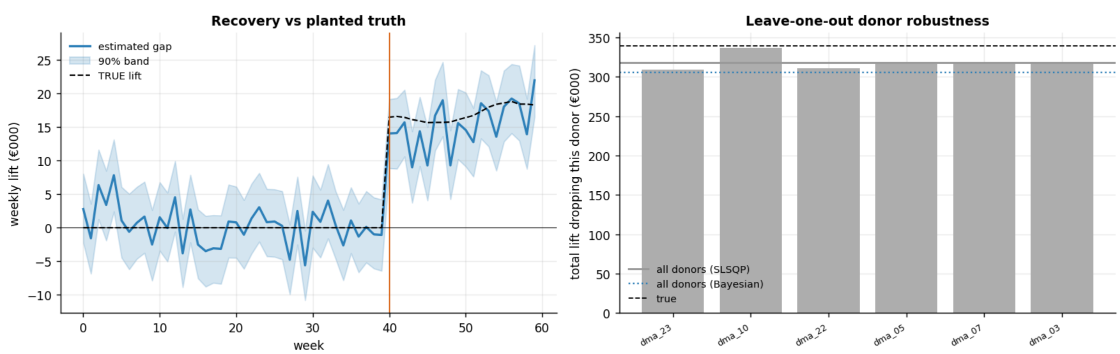
Four checks — and one of them delivers a genuinely humbling result we keep rather than hide:

1. **Recovery & calibration.** The true per-week path should sit inside the posterior band. That band is a **posterior predictive**: it carries not only *which-donors* (weight) uncertainty

but the treated market's own idiosyncratic noise. That noise term is what makes the per-week band near-nominal; a weights-only band is too narrow and can leave the truth *outside* a "90%" interval.

2. **Multi-seed calibration.** One panel is a single draw of the world, so we refit on fresh panels and report the recovery bias and coverage. The honest result: the **per-week** band is near-nominal, but the **total-lift** interval covers well *below* nominal — synthetic control's genuine sample-to-sample variance is larger than any single panel's posterior width. So the posterior interval is an *optimistic* guide to the total, and the trustworthy inference is the **placebo permutation test** (design-based, calibrated by construction), not the band. This is *why* the euro call below is "confirm with a second test," not "the interval says go." *(The per-seed refits in this check are deliberately coarse 300-draw probes, read only in aggregate; their sampler chatter is silenced.)*
3. **Pre-fit gate** — the pre-period gap must hover around **zero**: concretely, pre-RMSE below about a third of the weekly lift the design must detect (printed as an explicit PASS/FAIL below). A bad synthetic match makes the post gap meaningless — this is the precondition, not a formality.
4. **Leave-one-out donor robustness** — drop each of the top-weighted donors and refit; if the estimate lurches when one donor leaves, the result hangs on a single market and is fragile.

```
pre-fit gate: pre-RMSE €2.86k  vs  |weekly lift|/3 = €5.1k  ->  PASS
pre-launch gap mean €0.34k (~0 = good fit).  LOO total range €309-337k vs the
same-fitter SLSQP baseline €318k (Bayesian €305k) - no single donor drives it.
```




```
Across 10 fresh panels - recovery bias €+38k (+11%), with a LARGE seed-to-seed
sd of €56k.
PER-WEEK band covers the true path 87% of the time - ~ nominal; the posterior-
predictive obs-noise term earns this.
But the TOTAL-lift 90% interval covers only ~40% across panels: SC's sample-to-
sample variance dwarfs any single panel's posterior width, so the total interval
is optimistic.
=> the CALIBRATED inference is the placebo permutation test (below), not the
posterior band alone.
```

Why does the total-lift band under-cover while the per-week bands are fine? This is the notebook's most important negative result, and it is not a mystery — it follows from one modelling choice. The likelihood treats the weekly errors as independent, so if e_t is the error of the synthetic blend in week t, the model prices the uncertainty of the $n = 20$-week **total** as

$$\mathrm{sd}\Big(\sum_{t\in\mathrm{post}} e_t\Big) \;=\; \sqrt{n}\,\sigma \qquad \text{(iid errors).}$$

But the blend's true error is *not* iid. Whatever part of the treated market's factor loadings the weights fail to match leaks the shared factors straight into the gap,

$$e_t \;\approx\; \big(\lambda_0 - \textstyle\sum_j w_j\lambda_j\big)^{\top}(g_t,\, s_t,\, m_t) \;+\; \varepsilon_{0t},$$

and m_t is a **random walk** — so the mismatch term is a *persistent, slowly wandering offset*, not fresh noise each week. For correlated errors the variance of a sum picks up all the covariance terms,

$$\mathrm{Var}\Big(\sum_{t=1}^{n} e_t\Big) \;=\; \sigma^2\Big(n \;+\; 2\sum_{k=1}^{n-1}(n-k)\,\rho_k\Big) \;\xrightarrow{\rho_k\to 1}\; n^2\sigma^2 \;\gg\; n\,\sigma^2,$$

where ρ_k is the error's lag-k autocorrelation. **Per-week bands only need the marginal variance** σ^2, which the posterior-predictive ε_t term gets roughly right — that is why they cover ~ nominal. **The total needs the covariances too**, and the iid likelihood sets every ρ_k to zero. The cell below measures both faces of this on live data: the *in-sample* pre-period residual shows only modest, sign-flipping autocorrelation (the fit whitens most of what it can see — which is why eyeballing the pre-fit residual would *not* warn you), while the *out-of-sample* gap error — measured on placebo worlds where the true lift is zero, so the post gap **is** the error — is almost perfectly persistent from week to week.

```
posterior sd of the 20-week total (mean across panels): €24k   vs   actual seed-
to-seed sd of the total error: €56k  ->  ~2.3x underestimated
in-sample pre-period residual ACF (lags 1-3): [-0.03  0.38 -0.12] - modest and
sign-flipping (noise band ~ +/-0.32); the fit whitens most of what it can see
out-of-sample gap-error autocorrelation about 0, placebo worlds (lags
1,2,3,5,8): [0.94 0.94 0.93 0.95 0.95]
Symptom: the total-lift band under-covers (above). Mechanism: the blend's
error is a PERSISTENT offset - the unmatched share of the macro random walk -
not fresh weekly noise, so the variance of the 20-week SUM scales like n^2,
not the n the iid likelihood charges for. Per-week bands only price the
marginal variance, which is why they stay ~ nominal while the total band is
optimistic. Lesson: iid likelihoods and aggregated-over-time estimands don't
mix - for totals, use design-based placebo inference (Depth B) or a
likelihood with correlated (AR / random-walk) errors.
```

0.5 6 · Decide, in euros

The campaign cost **€300k**. Everything the notebook has produced so far now collapses into a single object: the **posterior over total incremental sales** across the 20 post-launch weeks (`total_post`, in €000). The rollout question is *not* "is the lift positive?" — Depth B's placebo permutation answers that — but **"what is the probability the lift beats what we paid?"**, because a real-but-too-small effect still loses money at national scale.

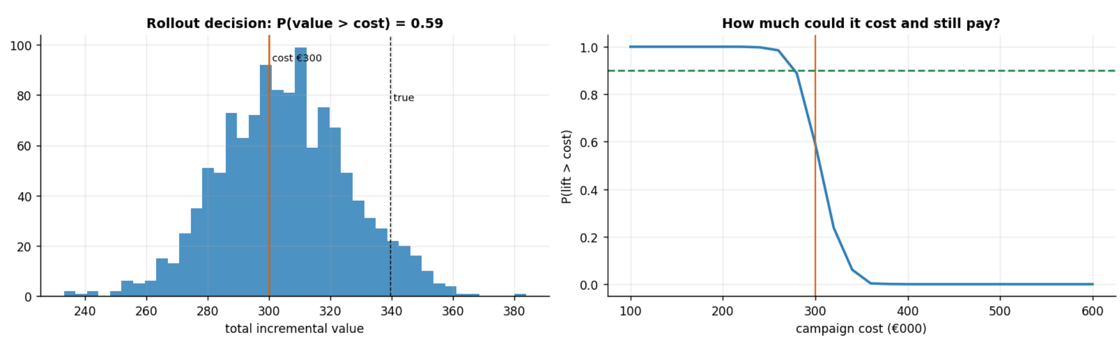
The decision rule (`policy.go_no_go`, the cookbook's standard): read $P(\text{value} > \text{cost})$ off the posterior and act on a stated convention —

- $P > 0.9$ -> **GO** — sure enough to act without buying more evidence;
- $P < 0.5$ -> **NO-GO** — more likely than not the campaign doesn't pay at this price;
- in between -> **TEST FURTHER** — the honest middle where the effect may well be real but the *economics* are not yet settled.

Alongside the probability it reports the expected **net value** ($\mathbb{E}[\text{value}] - \text{cost}$, in €000) and the expected **ROI** as a ratio (net ÷ cost, so break-even is 0) — three views of the same posterior, so nobody has to reverse-engineer the call from a chart.

How to read the two panels. *Left*: the posterior of total lift with the €300k cost line — the mass to the right of the line **is** $P(\text{lift} > \text{cost})$ (the dashed marker is the true total, which we only know because this is a simulation). *Right*: the same posterior read as a **headroom curve** — sweep a hypothetical cost c and plot $P(\text{lift} > c)$. Where the curve crosses the green 0.9 line is the *highest price at which this evidence would already justify a confident GO* — the number to bring to the media-buying negotiation, not just a yes/no.


```
{
  "P_value_gt_cost": 0.59,
  "expected_value": 305.28,
  "expected_net_value": 5.28,
  "expected_roi": 0.02,
  "value_lo": 272.42,
  "value_hi": 341.54,
  "cost": 300.0,
  "decision": "TEST FURTHER"
}
```

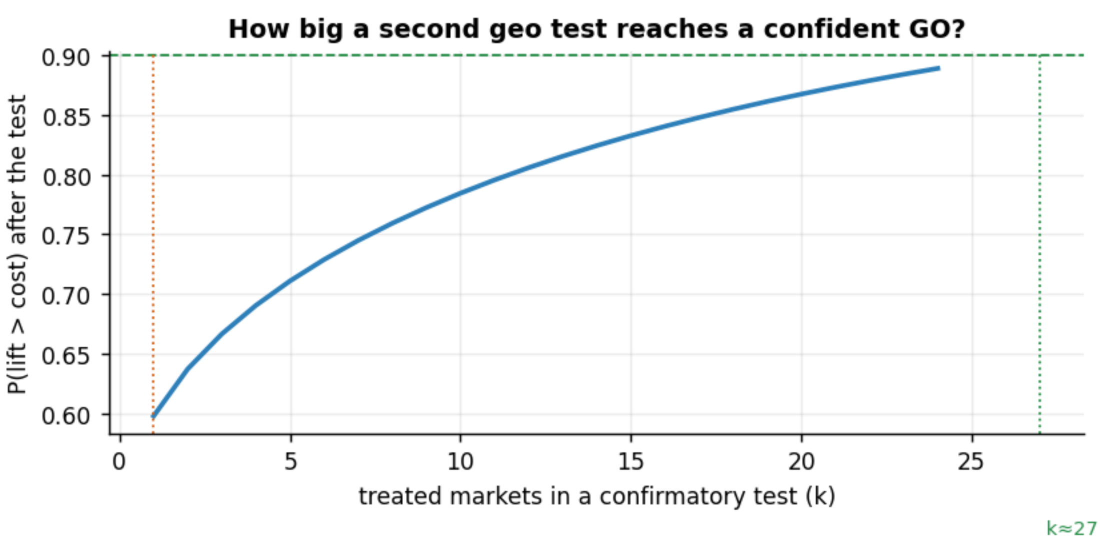



Read-out. The verdict in this run is **TEST FURTHER**, and the JSON says exactly why: the expected lift (~ €305k) sits almost on top of the €300k cost — expected net value ~ **€5k**, ROI ~ **+1.6%** — and $P(\text{lift} > \text{cost}) \approx 0.59$, barely better than a coin flip. Hold this against Depth B without flinching at the apparent contradiction: the placebo permutation says the effect is **real** (p ~ 0.03–0.05), while the economics say it is **marginal at this price**. Both are true at once — "statistically significant" and "worth the money" are different questions, and conflating them is how marginal campaigns get scaled. The headroom curve prices the gap: the 0.9 crossing sits at ~ €275k in this run, so had the campaign cost ~8% less, this same evidence would already be a confident GO. At €300k, the evidence buys conviction that *something* happened, not conviction that it *paid*.

A borderline call has a price tag: the value of information. "TEST FURTHER" must not end as a shrug — the deliverable is an answer to *"what evidence would settle this, and is it worth buying?"* Two tools, both computed below:

- **VOI (value of information).** $\text{VOI} = \mathbb{E}\big[\max(V-c,\,0)\big] - \max\big(\mathbb{E}[V]-c,\,0\big)$ — the expected gain from deciding *after* the uncertainty resolves versus committing now on the mean. Intuition: if the truth lands below the cost line, perfect foresight would have let us walk away and keep the difference; VOI is that averted loss, averaged over the posterior. It is the most a *perfectly informative* test could add in expected decision value — i.e. an **upper bound on any confirmatory-test budget**.
- **Test sizing.** A second geo test treating k independent markets tightens the posterior roughly as $1/\sqrt{k}$ — it **sharpens** the estimate around the same mean; it doesn't move it. So "how big must the test be?" becomes: *the smallest k whose tightened posterior reaches* $P(\text{lift} > \text{cost}) \geq 0.9$. Two hidden assumptions live inside that $1/\sqrt{k}$, and both flatter the test: it treats the k market-level estimates as **independent** with equal variance — but this whole world is built on *shared* macro factors, so the k gap estimates share estimation error and the *effective* k is smaller than the nominal one — and it takes this posterior's sd as the per-market scale, which the multi-seed check just showed runs optimistic. Both push the same way: read the computed k as a **lower bound**.

```
Value of information: resolving the cost-line uncertainty is worth ~€6k of
expected decision value.
A confirmatory geo test averaging ~27 independent treated markets would tighten
the interval
enough to reach P(lift>cost) >= 0.90 - and since the multi-seed check showed
this interval is
*optimistic*, treat 27 as a LOWER bound on the test size. That is the experiment
to run, not a blind launch.
```



Read-out — and an honest dead end. VOI ~ **€6k** in this run: with the posterior centred almost exactly on the cost line, even perfect information barely changes the expected outcome — whichever way the truth falls, we're near break-even, so the *stakes of deciding wrong are small.* That immediately disciplines the next step: a confirmatory test is only rational if it costs less than ~€6k of all-in decision value — no €50k research program is justified by these economics. The sizing arithmetic is blunter still: reaching the 0.9 bar would take ~ **33 independent treated markets, more than this world contains** (30 DMAs, one already treated). So no feasible single geo test settles the question at this cost. The honest options, in order: **renegotiate the price** (at ~ €275k the existing evidence already clears 0.9), **extend the post-period or pool repeated tests** (more weeks per market buys precision too), or **accept a lower confidence bar** knowing the multi-seed check in step 5 says these intervals are already optimistic. What is *not* on the list: a blind national rollout. Depth C pulls this thread together with the launch-date and donor-pool sensitivities into the one-paragraph verdict a VP signs.

0.6 7 · Caveats

- **Here synthetic control is high-variance, not conservative.** In this panel the point estimate lands ~10% low, but across seeds (Step 5) it shows only a small mean bias (+7% in our run) but a *large* sample-to-sample spread — and its posterior interval is *optimistic* about the total lift. So trust the **placebo permutation test**, not the point estimate or the band — §5 shows the mechanism: the blend's error is a *persistent* (serially correlated) offset, so the iid likelihood underprices the variance of any multi-week total.
- **Pre-fit is a gate, not a formality.** High pre-RMSE => the post gap is uninterpretable; no Bayesian machinery fixes a bad donor pool.
- **No-anticipation and no-spillover are assumptions.** Anticipation is stress-tested by placebo-in-time in Depth B; spillover is *invisible to every placebo*, so Depth B stress-tests it by simulation instead — planted spillover **attenuates** the estimate (SC under-states, never inflates), and the fix is donor-pool design, not statistics.
- **One treated unit = limited power.** Significance comes from the permutation test, coarse with few donors.

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1 Depth A · Why the naive estimators fail

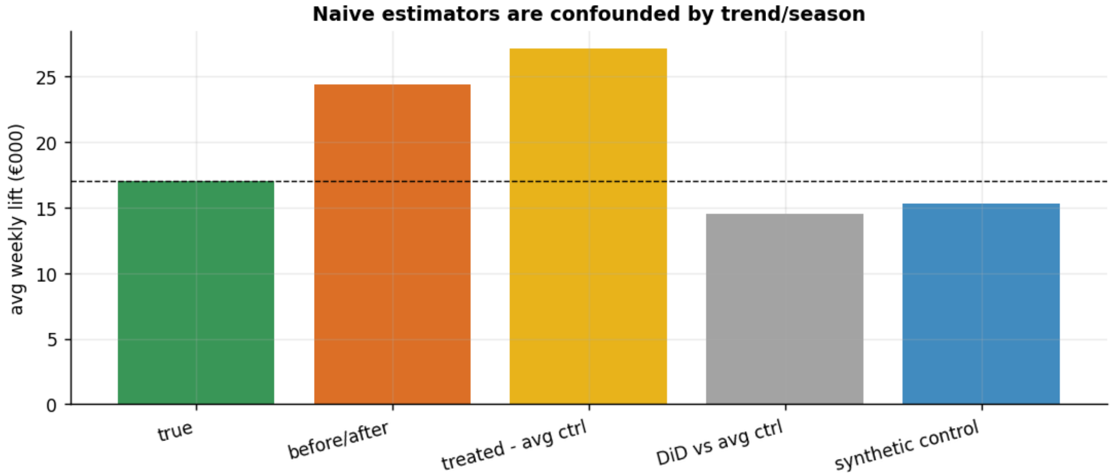
Two tempting shortcuts, both confounded by the shared trend/season/macro wave:

- **Before/after**: post mean − pre mean of the treated DMA — books the entire trend as effect.
- **Treated − average control**: the average control has the wrong factor loadings, so it doesn't cancel the common wave cleanly.

Synthetic control is the fix: a *weighted* control chosen to match pre-launch.


```
{
  "true": 17.0,
  "before/after": 24.3,
```

```
  "treated - avg ctrl": 27.1,
  "DiD vs avg ctrl": 14.5,
  "synthetic control": 15.3
}
```



Read-out. Before/after badly overstates the lift (it books the trend); treated-minus-average is **worse** — it books the raw between-market level gap, differencing nothing out; differencing out the shared trend (DiD vs the average control) removes the trend bias yet a *level* gap from mismatched factor loadings remains; **the reweighted synthetic control comes closest** (€15.2k vs the true €17.0k — an 11% undershoot; DiD-vs-average lands at €14.5k, and neither naive shortcut is even close). Same data, four estimates — the estimator *is* the identification strategy.

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2 Depth B · Placebo inference — is the gap statistically real?

We have a lift estimate, but with **one treated market** there's no classical standard error to test it (standard errors assume many independent units). Instead we ask a **permutation** question, the inferential heart of synthetic control:

> *If the campaign had no effect, how unusual would a post-launch gap this big be, just by chance?*

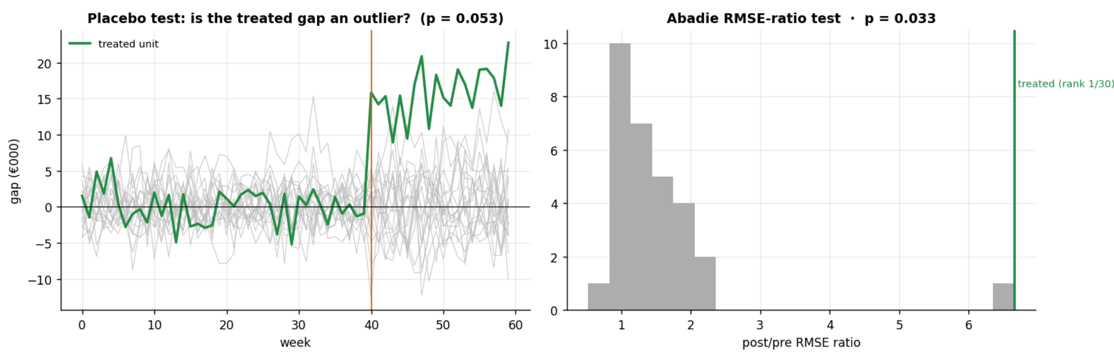
Placebo-in-space. We answer it by pretending, in turn, that *each donor market* was the treated one: fit that donor its own synthetic control from the remaining donors, and record its post-launch "gap." Since none of the donors actually got a campaign, these placebo gaps map out the distribution of gaps we'd see from noise alone. If the *real* treated market's gap sits far out in the tail of that placebo distribution, a fluke is unlikely. Two refinements from Abadie:

- **Discard placebos with a poor pre-fit** — a donor the synthetic can't match before launch produces a meaningless gap that would pollute the comparison.
- **Use the RMSE-ratio statistic**, not the raw gap: the *post*-period RMSE divided by the *pre*-period RMSE. This is **scale-free** (fair across big and small markets) and, crucially, doesn't

reward a placebo that simply fits badly *everywhere* (high post *and* high pre -> ratio ~ 1).

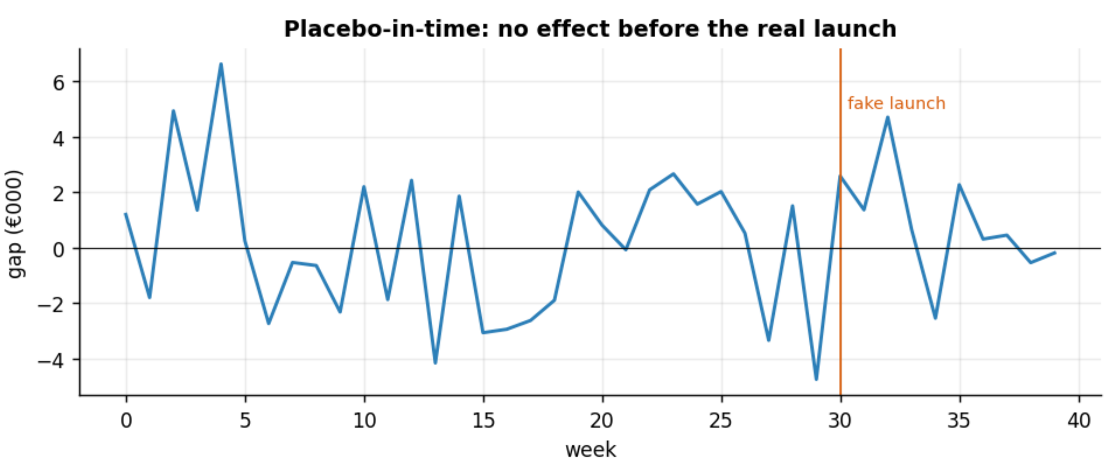
The **p-value** is then the treated market's rank in the ratio distribution: the fraction of markets (placebos + treated) whose ratio is at least as extreme as the treated market's. A small p — say <= 0.05 — means "only this market, out of all of them, shows a jump this sharp relative to its own pre-fit."

```
placebo-in-space kept 18/29 donors (dropped 11: pre-RMSE > 5x the treated fit)
placebo p vs filter cutoff: 2x -> p=0.056  ·  5x -> p=0.053  ·  20x -> p=0.333
(tighter filters compare only well-matched placebos and sharpen the test; a
loose 20x filter admits poor synthetics and dilutes it)
placebo p (gap) = 0.053 [add-one permutation, treated excluded from placebos]  ·
RMSE-ratio p = 0.033 [Abadie rank/N, treated included]  (treated ratio 6.7 vs
placebo median 1.3; both rank the treated 1st)
```



Placebo-in-time. A second falsification: pretend the campaign launched *before* it did (a fake week inside the pre-period). The estimated "effect" should be ~ 0 — a non-zero fake effect would mean the method invents lift (anticipation or overfitting).

```
placebo-in-time avg gap €0.9k  vs  pre-fit RMSE €2.86k (should be ~ 0, and it is
well within the pre-fit noise) - no spurious pre-effect.
```



Spillover — stress-testing the one assumption the placebos cannot touch. The assumption table in §3 was honest that placebo-in-space *cannot* detect spillover: if the campaign leaks into donor markets, every placebo refit is contaminated in exactly the same way, so nothing looks unusual. But the simulator can do what no real dataset can — rerun the same world with spillover *planted*. Below, a fraction φ of the true lift Δ_t leaks into five "neighbouring" donor markets ($Y_{jt} \mathrel{+}= \varphi\,\Delta_t$ for j in the neighbour set) and we refit. First-order prediction *before* running it: the contaminated donors rise, so the synthetic counterfactual rises, so the measured gap **shrinks** by roughly the spilled weight,

$$\text{bias} \;\approx\; -\varphi \sum_{j\in\text{neighbours}} w_j \;\times\; \text{lift},$$

i.e. spillover makes synthetic control **under-state** a real campaign — attenuation toward "no effect", the opposite of the false-positive failure most people fear.

```
 phi  est_total_k  weight_on_neighbours
0.00       317.74                  0.15
0.25       304.75                  0.15
0.50       291.75                  0.15    (true total €339k)
attenuation at phi=0.5: measured €-26k vs first-order prediction -phi·(Sigma
neighbour weights)·lift = €-24k
excluding the 5 neighbours at phi=0.5 restores the estimate to €311k
(contaminated pool: €292k; clean pool: €318k)
Read-out: spillover into donors raises the synthetic, so SC UNDER-states the
lift - attenuation, not inflation. Conservative for a GO call, but it can
turn a true GO into a NO-GO. And because no placebo can flag it, the fix is
DESIGN, not statistics: exclude adjacent / media-overlapping donors up
front - as just shown, a clean-but-smaller pool beats a bigger contaminated one.
```

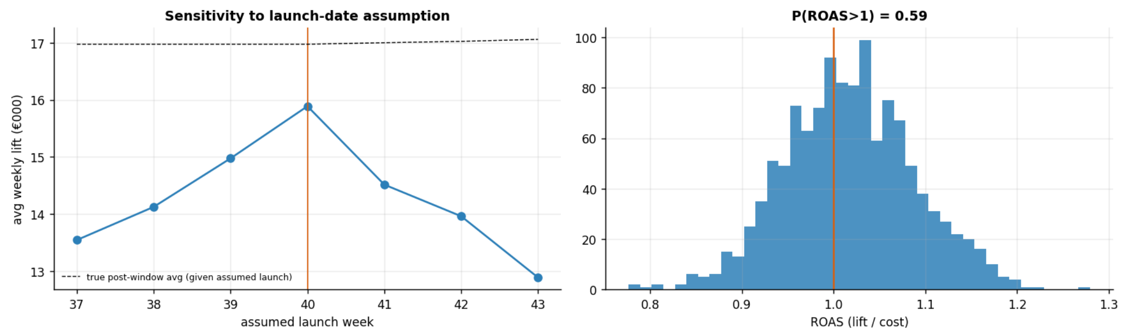
---

3 Depth C · The euro rollout decision & sensitivity

Pull it together into the number a VP signs off on, and pressure-test how the call moves with the **launch-timing** assumption (did we mis-date the campaign?) and the **donor pool**.


```
{
  "true_total": 339.48,
  "estimated_total": 305.28,
  "ci90": [
    272.42,
    341.54
  ],
  "pre_rmse": 2.86,
  "p_space": 0.05,
  "p_rmse_ratio": 0.03,
  "placebo_in_time": 0.9,
```

```
  "P_lift_gt_cost": 0.59,
  "decision": "TEST FURTHER",
  "eff_donors": 14.68
}
```

3.0.1 The one-paragraph decision

The campaign drove a **real** incremental lift: the synthetic tracks the treated market closely before launch (low pre-RMSE, pre-gap ~ 0), spreads its weight across ~15 effective donors (no single market dominates), and it is robust to dropping any single donor (leave-one-out stays close to the same-fitter baseline). Nudging the *assumed* launch date does shift the estimate — but that is mechanical, not fragility: mis-dating folds no-effect weeks into the post window and dilutes the average lift, so the peak at the true launch week is itself a diagnostic. The post gap is an outlier on both the raw-gap and the scale-free RMSE-ratio **placebo permutation** tests, and the time placebo shows no spurious pre-effect — that permutation test is the calibrated evidence the effect is real. The euro call is deliberately cautious: the point estimate carries meaningful seed-to-seed variance (this panel lands ~10% below truth; across panels the mean bias is small (+7%) but the variance is *large*), and the multi-seed check shows the **total-lift interval is optimistic** — so we do **not** let a single posterior interval authorise a national rollout. Against the €300k spend the effect sits near break-even: **"promising, confirm with a second geo test," not "scale nationally today."** The effect is clearly real; whether it clears *this* cost is the close call — and one an optimistic interval should make us *more*, not less, cautious about.